



Basic Details of the Team and Problem Statement

Ministry/Organization Name/Student Innovation:
Ministry of Ports, Shipping and Waterways

PS Code: SIH1310

Problem Statement Title: Drone-based surveillance system for the vessels plying in port areas and encroachments

Team Name: High Flyers

Team Leader Name: Vaibhav Tiwari

Institute Code (AISHE): U-0385

Institute Name: Thapar Institute of Engineering & Technology

Theme Name: Robotics and Drones

IDEA AND TECH TO BE USED



BRIEF INTRODUCTION TO OUR SOLUTION

The **Smart Port Management System** with Drone Integration offers a cost-effective solution to track vessels and detect encroachments in the port.



TRACKING AND ENCROACHMENT SYSTEM

A fleet of **camera-enabled drones** coordinate to continuously monitor vessel movements within the port's defined limits. Using **geo-fencing**, we establish boundaries and cross-reference vessel positions obtained from the drone feed with AIS data and port schedule. Any mismatch **triggers alerts** to the port administration. This information is displayed in a user-friendly dashboard.

Machine Learning algorithms are employed to track vessel movement and detect any **suspicious activity**, as well as to detect and monitor unverified vessels around the port limits.



MMIS CENTRALIZATION AND INTEGRATION

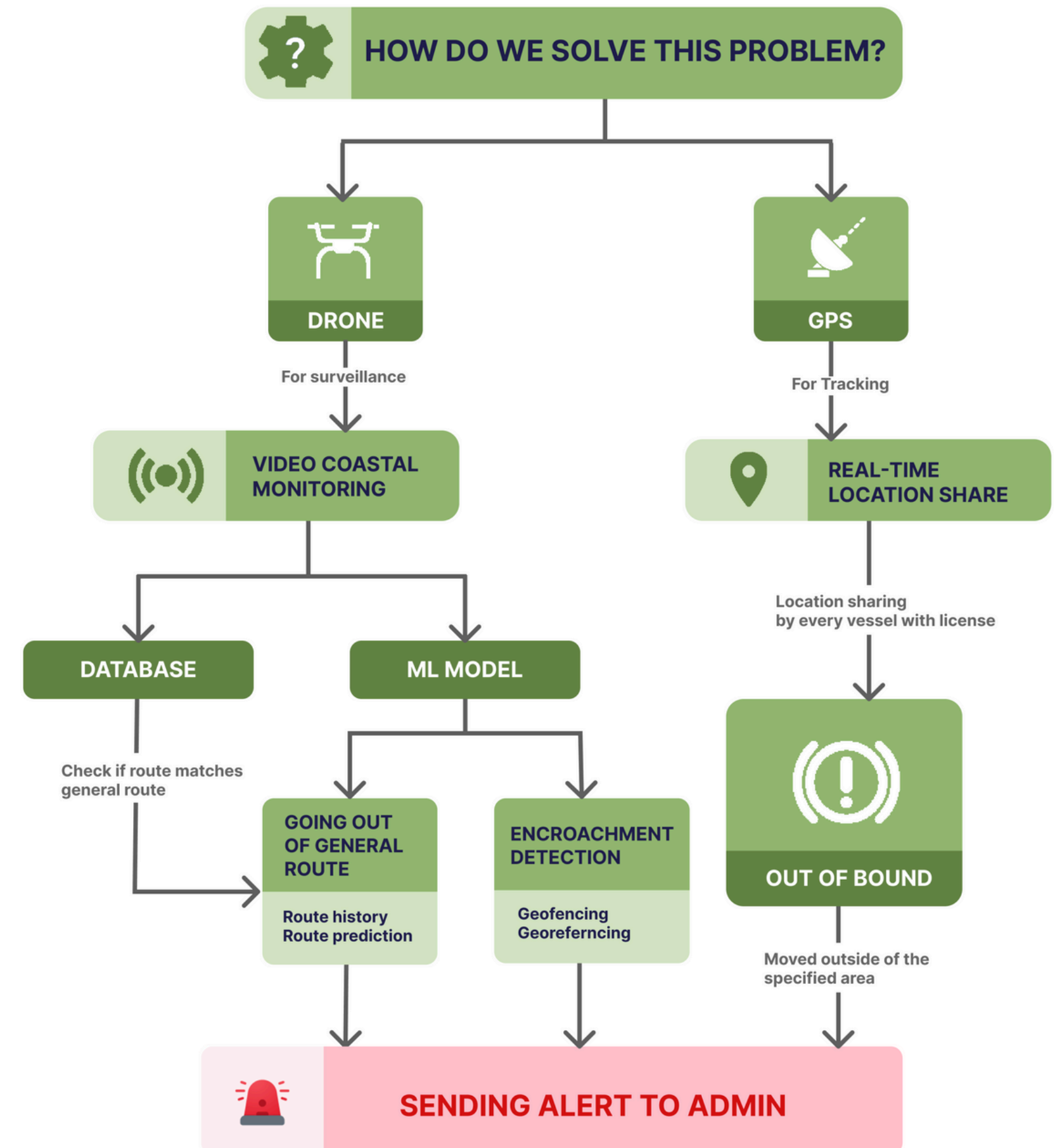
Centralized repository for violator, encroachment, vessel activity, and security incident data to continuously improve ML models and port security.



TECHNOLOGICAL STACK



FLOW DIAGRAM



SHOW STOPPERS AND USE CASES



PROMPT ALERT SYSTEM

Verify ships in real-time and send an **alert** to the authorities in case an intruder is detected, as well as **actively track** that vessel using drones.



AUTONOMOUS DRONE PROGRAMMING

Manages take-off, monitoring, and return to base procedures, effectively mitigating **blind spots in surveillance**. Sequential drone deployment minimizes monitoring gaps.



ADVANCED DEEP LEARNING

Utilizing state-of-the-art deep learning technology for precise ship recognition, velocity tracking, and **trajectory prediction**, facilitating accurate and proactive decision-making.

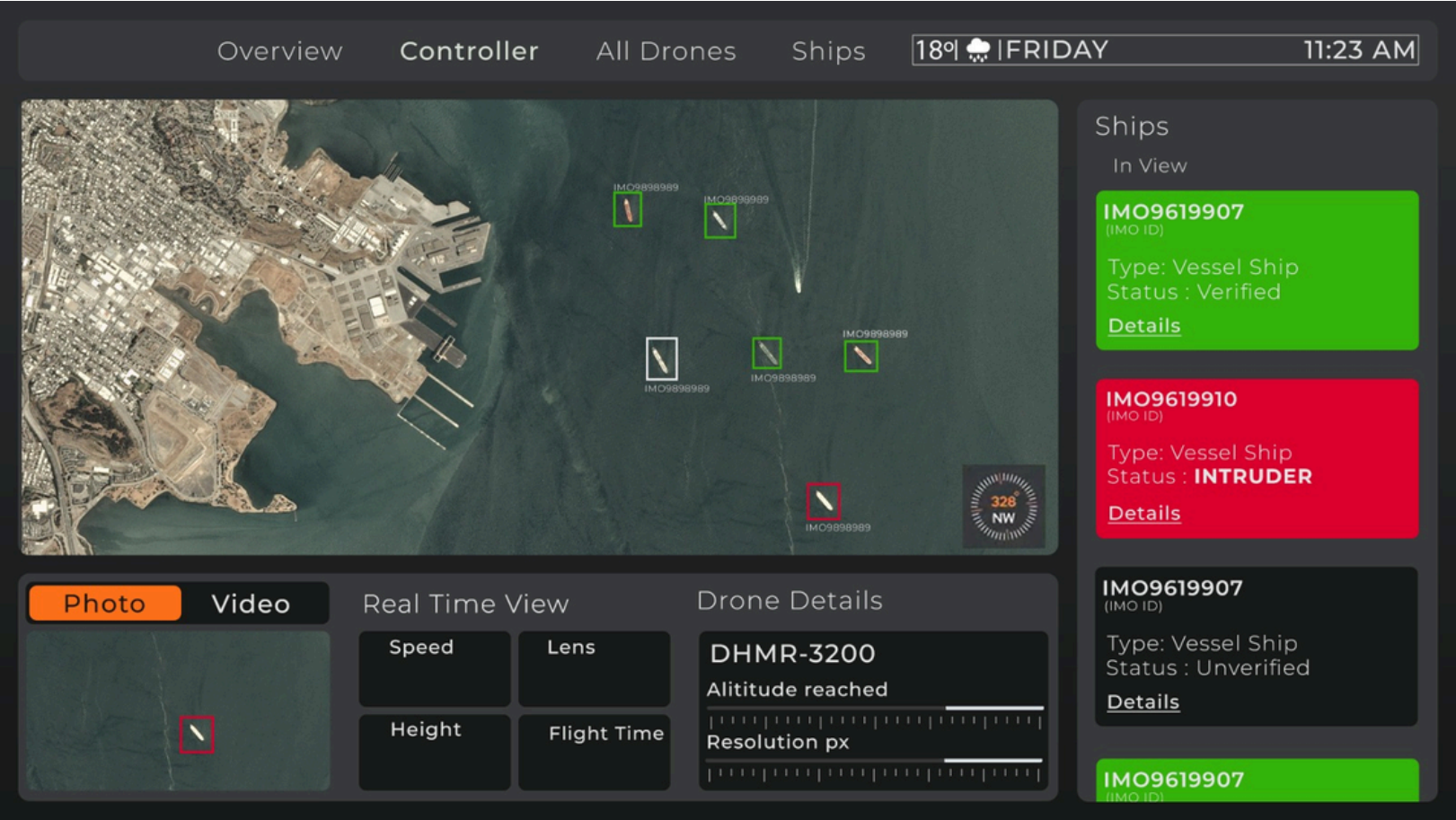
DRONE

- Ship Detection and Verification
- Real Time tracking
- AIS Integration
- Geo-Referencing

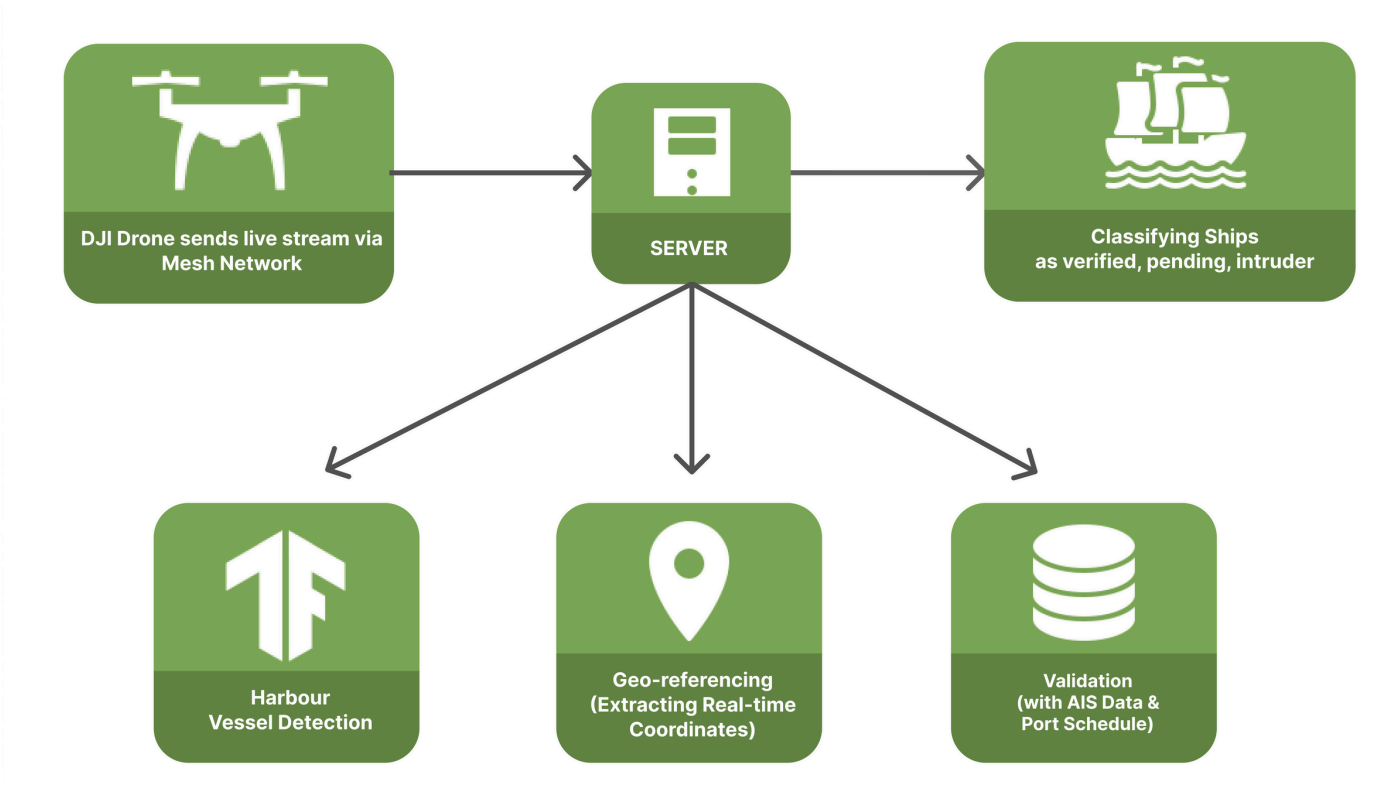
ADMIN

- Surveillance
- Real time alerts
- Manage drones efficiently to get more coverage

PROTOTYPE



Allowing real-time monitoring and surveillance within the port seamlessly.



Procedural verification of vessels

Team Member Details

Team Leader Name: Vaibhav Tiwari

Branch: BE Stream: CSE Year: III

Team Member 1 Name: Kush Mehta

Branch: BE Stream: CSE Year: III

Team Member 2 Name: Dhruv Lotia

Branch: BE Stream: CSE Year: III

Team Member 3 Name: Uddeepta Raaj Kashyap

Branch: BE Stream: CSE Year: III

Team Member 4 Name: Arushi Dhingra

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Team Member 5 Name: Satyajeet Bedi

Branch: BE Stream: CSE Year: III

